

Module 10 - chapter 24

Demo project: Deploy Microservices with Helmfile

Continuing from Module 10 -chapter 23 we have the following charts:

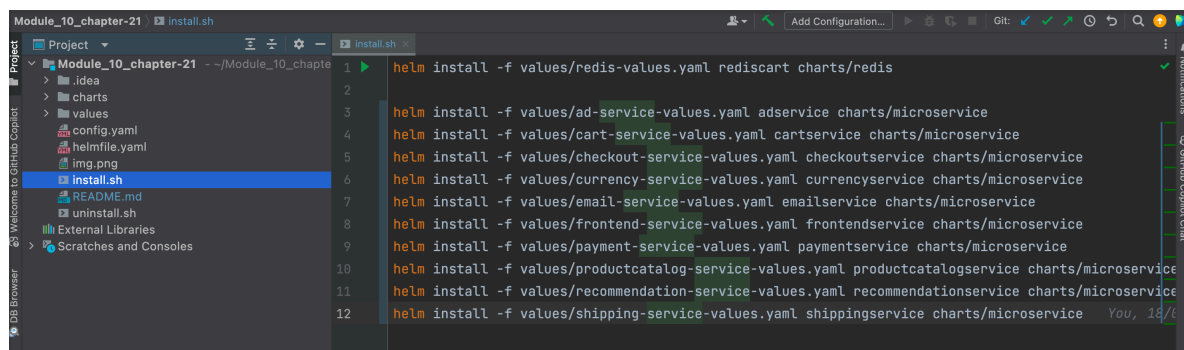
- Redis
- MS chart (10 apps)

We are ready to deploy them to our k8s cluster, we have two option to do that:

1. One by one using

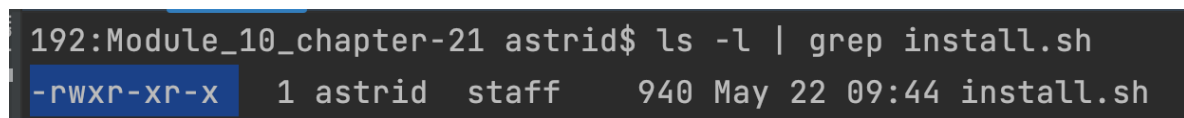
-> `helm install -f values/redis-values.yaml rediscart charts/redis`

We can do it manually in th terminal or we can write a shell script:



```
1 helm install -f values/redis-values.yaml rediscart charts/redis
2
3 helm install -f values/ad-service-values.yaml adservice charts/microservice
4 helm install -f values/cart-service-values.yaml cartservice charts/microservice
5 helm install -f values/checkout-service-values.yaml checkoutservice charts/microservice
6 helm install -f values/currency-service-values.yaml currencyservice charts/microservice
7 helm install -f values/email-service-values.yaml emailservice charts/microservice
8 helm install -f values/frontend-service-values.yaml frontend-service charts/microservice
9 helm install -f values/payment-service-values.yaml paymentservice charts/microservice
10 helm install -f values/productcatalog-service-values.yaml productcatalogservice charts/microservice
11 helm install -f values/recommendation-service-values.yaml recommendationservice charts/microservice
12 helm install -f values/shipping-service-values.yaml shippingservice charts/microservice
```

Lets check if the file executable:



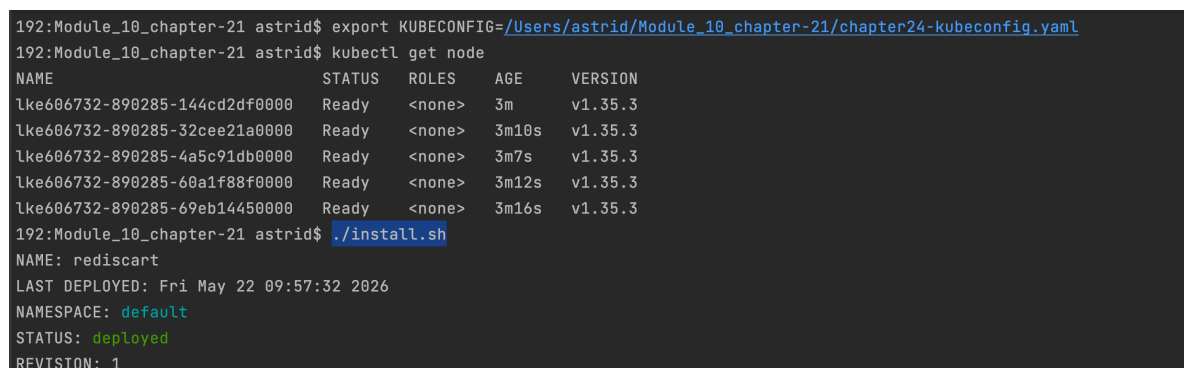
```
192:Module_10_chapter-21 astrid$ ls -l | grep install.sh
-rwxr-xr-x  1 astrid  staff   940 May 22 09:44 install.sh
```

It has the X -> already executable. In case it wasn't then :

-> `chmod u+x install.sh`

Now we run the script

-> `./install.sh`



```
192:Module_10_chapter-21 astrid$ export KUBECONFIG=/Users/astrid/Module_10_chapter-21/chapter24-kubeconfig.yaml
192:Module_10_chapter-21 astrid$ kubectl get node
NAME                                STATUS    ROLES    AGE    VERSION
lke606732-890285-144cd2df0000      Ready    <none>    3m     v1.35.3
lke606732-890285-32cee21a0000      Ready    <none>    3m10s  v1.35.3
lke606732-890285-4a5c91db0000      Ready    <none>    3m7s   v1.35.3
lke606732-890285-60a1f88f0000      Ready    <none>    3m12s  v1.35.3
lke606732-890285-69eb14450000      Ready    <none>    3m16s  v1.35.3
192:Module_10_chapter-21 astrid$ ./install.sh
NAME: rediscart
LAST DEPLOYED: Fri May 22 09:57:32 2026
NAMESPACE: default
STATUS: deployed
REVISION: 1
```

-> `kubectl get pod`

```
Terminal: Local x + v
192:Module_10_chapter-21 astrid$ kubectl get pod
NAME                                READY   STATUS    RESTARTS   AGE
adservice-5f8b8889c9-8st59          1/1     Running   0           82s
adservice-5f8b8889c9-qwx4f          1/1     Running   0           82s
cartservice-548cc667f7-fjckv        1/1     Running   0           82s
cartservice-548cc667f7-pgqbh        1/1     Running   0           82s
checkoutservice-d5cdb7ffd-9xzf9      1/1     Running   0           82s
checkoutservice-d5cdb7ffd-swn7z      1/1     Running   0           82s
currencyservice-dbb475f87-2jgw8      1/1     Running   0           82s
currencyservice-dbb475f87-jkv29      1/1     Running   0           82s
emailservice-778b6cfbdd-4pjmj        1/1     Running   0           81s
emailservice-778b6cfbdd-qk2g6        1/1     Running   0           81s
frontend-787f799d5f-cz8kn           1/1     Running   0           81s
frontend-787f799d5f-m6jnp           1/1     Running   0           81s
paymentservice-85566bc778-2pzkcn     1/1     Running   0           81s
paymentservice-85566bc778-nq45h      1/1     Running   0           81s
productcatalogservice-549b86c956-55dr4 1/1     Running   0           81s
productcatalogservice-549b86c956-9cxr7 1/1     Running   0           81s
recommendationservice-6b74f8cb9f-nm4m2 1/1     Running   0           80s
recommendationservice-6b74f8cb9f-scjm9 1/1     Running   0           80s
redis-cart-9795bb64c-2jwc7           1/1     Running   0           83s
redis-cart-9795bb64c-xvxs           1/1     Running   0           83s
shippingservice-56c958dbf6-9v5rv      1/1     Running   0           80s
shippingservice-56c958dbf6-kkb4q      1/1     Running   0           80s
192:Module_10_chapter-21 astrid$
```

But this way is not elegant and not efficient 😞

Now to uninstall we have to do it for each of the helm releases 😞

```
Module_10_chapter-21 - 18/05/2026
install.sh x uninstall.sh x
1 values.yaml redis-cart charts/redis 1 helm uninstall redis-cart
2 2
3 ice-values.yaml adservice charts/microservice 3 helm uninstall adservice
4 vice-values.yaml cartservice charts/microservice 4 helm uninstall cartservice
5 :-service-values.yaml checkoutservice charts/microservice 5 helm uninstall checkoutservice
6 /-service-values.yaml currencyservice charts/microservice 6 helm uninstall currencyservice
7 rvice-values.yaml emailservice charts/microservice 7 helm uninstall emailservice
8 f-service-values.yaml frontend-service charts/microservice 8 helm uninstall frontend-service
9 -service-values.yaml paymentservice charts/microservice 9 helm uninstall paymentservice
10 catalog-service-values.yaml productcatalogservice charts/microservice 10 helm uninstall productcatalogservice
11 dation-service-values.yaml recommendationservice charts/microservice 11 helm uninstall recommendationservice
12 g-service-values.yaml shippingservice charts/microservice You, 18/05/2026, 12 helm uninstall shippingservice
13
```

```
192:Module_10_chapter-21 astrid$ ls -l | grep uninstall.sh
-rwxr-xr-x 1 astrid staff 334 May 22 10:06 uninstall.sh
192:Module_10_chapter-21 astrid$ ./uninstall.sh
release "rediscart" uninstalled
release "adservice" uninstalled
release "cartservice" uninstalled
release "checkoutservice" uninstalled
release "currencyservice" uninstalled
release "emailservice" uninstalled
release "frontendservice" uninstalled
```

-> kubectl get pod

```
192:Module_10_chapter-21 astrid$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
currencyservice-dbb475f87-2jgw8	1/1	Terminating	0	9m22s
currencyservice-dbb475f87-jkv29	1/1	Terminating	0	9m22s
emailservice-778b6cfbdd-4pjmj	1/1	Terminating	0	9m21s
emailservice-778b6cfbdd-qk2g6	1/1	Terminating	0	9m21s
paymentservice-85566bc778-2pzkn	1/1	Terminating	0	9m21s
paymentservice-85566bc778-nq45h	1/1	Terminating	0	9m21s
recommendationservice-6b74f8cb9f-nm4m2	1/1	Terminating	0	9m20s
recommendationservice-6b74f8cb9f-scjm9	1/1	Terminating	0	9m20s

```
192:Module_10_chapter-21 astrid$
```

-> kubectl get pod

```
192:Module_10_chapter-21 astrid$ kubectl get pod
No resources found in default namespace.
```

No helm releases anymore:

-> helm ls

```
192:Module_10_chapter-21 astrid$ helm ls
```

NAME	NAMESPACE	REVISION	UPDATED	STATUS	CHART	APP VERSION
------	-----------	----------	---------	--------	-------	-------------

```
192:Module_10_chapter-21 astrid$
```

2. Use a helmfile -> more elegant 👍

What is a Helmfile?

Declarative way to deploy helm charts in a single yaml file. We define the desired state, by saying: I want these releases to run on my cluster and this is the configuration I want:

What is a Helmfile?

Declarative way for deploying helm charts

helmfile.yaml

Define the desired state!

```
releases:
- name: emailservice
  chart: ../charts/app
  values:
    - ../values/emailservice.yaml
- name: paymentservice
  chart: ../charts/app
  values:
    - ../values/paymentservice.yaml
.....
```

- Declare a definition of an entire K8s cluster
- Change specification depending on application or type of environment

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And we can change the specification depending on the application or the type of env like dev, QA, prod.

Below is the template to configure a list of releases in the helmfile.yaml:

```
releases:
- name: XXXX
  chart: XXXX
  values:
    - xxxx.yaml
    - appReplcas: "XXXX"
    - volumeName: "XXXX"
```

```
helm install -f values/redis-values.yaml redisca charts/redis
helm install -f values/ad-service-values.yaml adservice charts/micro
helm install -f values/cart-service-values.yaml cartservice charts/mi
helm install -f values/checkout-service-values.yaml checkoutservice
helm install -f values/currency-service-values.yaml currencyservice
helm install -f values/email-service-values.yaml emailservice charts
helm install -f values/frontend-service-values.yaml frontendservice
helm install -f values/payment-service-values.yaml paymentservice cha
helm install -f values/productcatalog-service-values.yaml productcata
helm install -f values/recommendation-service-values.yaml recommendat
helm install -f values/shipping-service-values.yaml shippingservice c
```

Let's do Redis first:

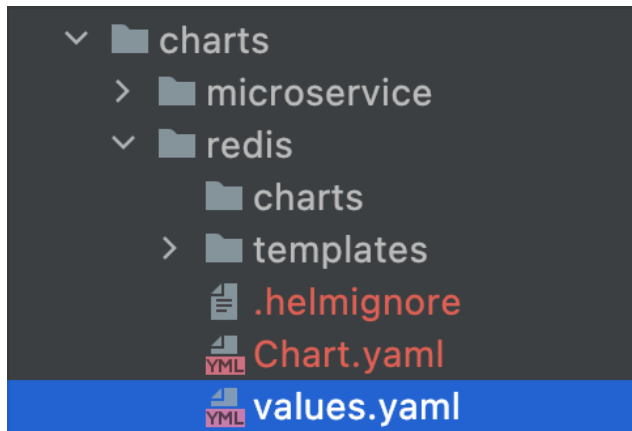
```
releases:
- name: redisca
  chart: charts/redis
  values:
    - values/redis-values.yaml
  # - appReplcas: "XXXX" # already configure in values/redis-values.yaml
  # - volumeName: "XXXX" # already configure in values/redis-values.yaml
```

```
helm install -f values/redis-values.yaml redisca
helm install -f values/ad-service-values.yaml adser
helm install -f values/cart-service-values.yaml car
helm install -f values/checkout-service-values.yaml
helm install -f values/currency-service-values.yaml
helm install -f values/email-service-values.yaml en
helm install -f values/frontend-service-values.yaml
```

If I uncomment lines 6 and 7 then they will override values/redis-values.yaml

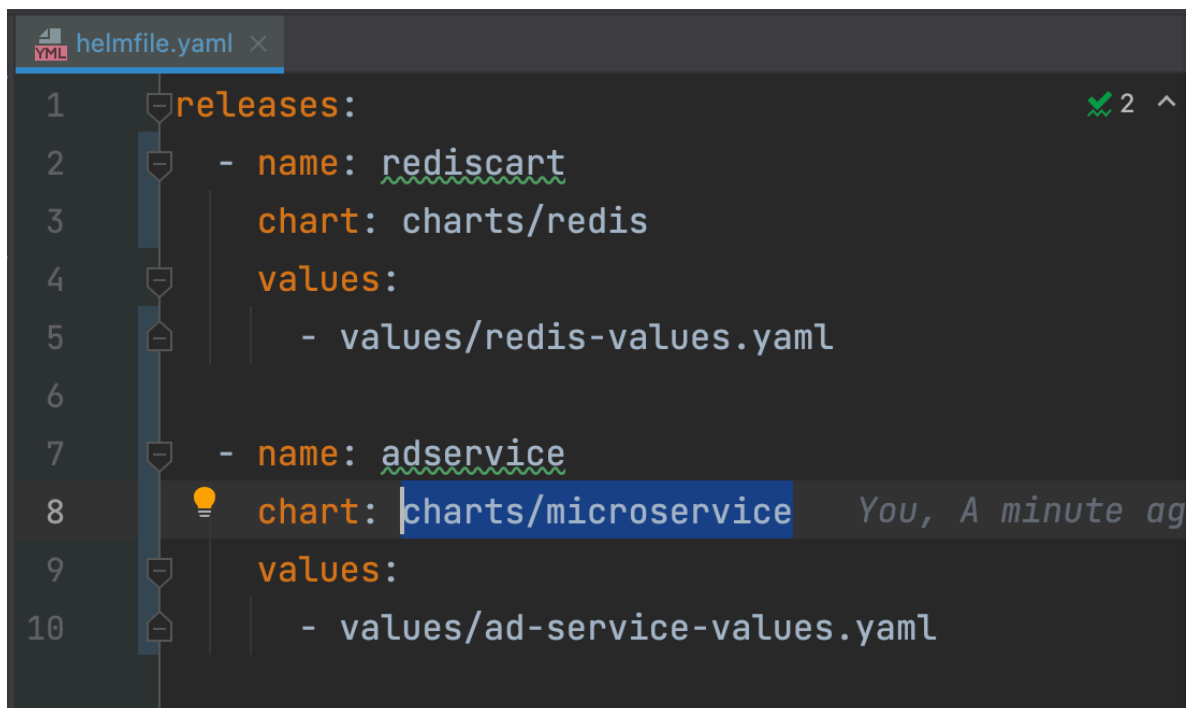
and values.yaml.

The default values in values.yaml inside the redis chart folder -> always is executed first:



Then the values/redis-values.yaml overrides the defaults from values.yaml.

Next -> adservice:



Installing the helmfile

I have added all of the releases to the helmfile, now we can install the helmfile:

- install the helmfile tool: -> brew install helmfile

Now let's execute the helmfile:

-> helmfile sync

Prepares the releases, screening the helmfile configuration:

```
192:Module_10_chapter-21 astrid$ helmfile sync
Building dependency release=rediscart, chart=charts/redis
Building dependency release=cartservice, chart=charts/microservice
Building dependency release=currencyservice, chart=charts/microservice
Building dependency release=emailservice, chart=charts/microservice
Building dependency release=checkoutservice, chart=charts/microservice
Building dependency release=frontendservice, chart=charts/microservice
Building dependency release=productcatalogservice, chart=charts/microservice
Building dependency release=paymentservice, chart=charts/microservice
Building dependency release=recommendationservice, chart=charts/microservice
Building dependency release=shippingservice, chart=charts/microservice
```

Then compares the current state of the cluster with the desired state provided by the helm configuration:

```
Upgrading release=paymentservice, chart=charts/microservice, namespace=
Upgrading release=productcatalogservice, chart=charts/microservice, namespace=
Release "cartservice" does not exist. Installing it now.
NAME: cartservice
LAST DEPLOYED: Fri May 22 10:49:23 2026
NAMESPACE: default
```

And then figures out what it needs to be done to get to the desired state, for example: 'Installing'

```
Listing releases matching ^checkoutservice$
Release "frontendservice" does not exist. Installing it now.
NAME: frontendservice
LAST DEPLOYED: Fri May 22 10:49:23 2026
NAMESPACE: default
STATUS: deployed
REVISION: 1
DESCRIPTION: Install complete
TEST SUITE: None
```

```
Listing releases matching ^frontendservice$
Release "rediscart" does not exist. Installing it now.
NAME: rediscart
LAST DEPLOYED: Fri May 22 10:49:23 2026
NAMESPACE: default
STATUS: deployed
REVISION: 1
DESCRIPTION: Install complete
TEST SUITE: None
```

frontendservice	default	1	2026-05-22 10:49:23.184827 +0100 BST	deployed	microservice-0.1.0	1.16.0
paymentservice	default	1	2026-05-22 10:49:23.09919 +0100 BST	deployed	microservice-0.1.0	1.16.0
checkoutservice	default	1	2026-05-22 10:49:23.386889 +0100 BST	deployed	microservice-0.1.0	1.16.0
productcatalogservice	default	1	2026-05-22 10:49:23.108882 +0100 BST	deployed	microservice-0.1.0	1.16.0
rediscart	default	1	2026-05-22 10:49:23.325646 +0100 BST	deployed	redis-0.1.0	1.16.0

It shows us the list of releases:

UPDATED RELEASES:

NAME	NAMESPACE	CHART	VERSION	DURATION
cartservice		charts/microservice	0.1.0	2s
adservice		charts/microservice	0.1.0	2s
paymentservice		charts/microservice	0.1.0	2s
recommendationservice		charts/microservice	0.1.0	2s
currencyservice		charts/microservice	0.1.0	2s
emailservice		charts/microservice	0.1.0	2s
productcatalogservice		charts/microservice	0.1.0	2s
shippingservice		charts/microservice	0.1.0	2s
checkoutservice		charts/microservice	0.1.0	2s
frontendservice		charts/microservice	0.1.0	2s
redis-cart		charts/redis	0.1.0	2s

We can also see the list of releases -> helmfile list

Now let's check the pods:

-> kubectl get pod

```
192:Module_10_chapter-21 astrid$ kubectl get pod
NAME                                READY   STATUS    RESTARTS   AGE
adservice-5f8b8889c9-n5kk7         1/1     Running   0           12m
adservice-5f8b8889c9-vnjsv         1/1     Running   0           12m
cartservice-548cc667f7-bhjnz       1/1     Running   0           12m
cartservice-548cc667f7-wj6jf       1/1     Running   0           12m
checkoutservice-d5cdb7ffd-p824x     1/1     Running   0           12m
checkoutservice-d5cdb7ffd-vmksn     1/1     Running   0           12m
currencyservice-dbb475f87-d74mv     1/1     Running   0           12m
currencyservice-dbb475f87-gwhpv     1/1     Running   0           12m
emailservice-778b6cfbdd-bbbpt       1/1     Running   0           12m
emailservice-778b6cfbdd-nsxpc       1/1     Running   0           12m
frontend-787f799d5f-g8ds5           1/1     Running   0           12m
frontend-787f799d5f-m6l5g           1/1     Running   0           12m
paymentservice-85566bc778-fp4gj     1/1     Running   0           12m
paymentservice-85566bc778-zh2m9     1/1     Running   0           12m
productcatalogservice-549b86c956-l4fmf 1/1     Running   0           12m
productcatalogservice-549b86c956-wlcvg 1/1     Running   0           12m
recommendationservice-6b74f8cb9f-fxt8q 1/1     Running   0           12m
recommendationservice-6b74f8cb9f-vzpgd 1/1     Running   0           12m
redis-cart-9795bb64c-nmprw          1/1     Running   0           12m
redis-cart-9795bb64c-wc4rL          1/1     Running   0           12m
shippingservice-56c958dbf6-99zL6     1/1     Running   0           12m
shippingservice-56c958dbf6-xgwm4     1/1     Running   0           12m
192:Module_10_chapter-21 astrid$
```

Now lets look at the browser:

NodeBalancers / lke606732-da4fddb9e921

Summary Configurations Metrics **NEW** Settings

Backend Status ● 5 Up - 0 Down

Type Basic	Configuration Ports 80
Region GB, London 2	Hostname 172-236-5-79.ip.linodeusercontent.com
NodeBalancer ID 2213879	Transferred 62.46 KB

And I got an error:

 **ONLINEBOUTIQUE** 

Uh, oh!

Something has failed. Below are some details for debugging.

HTTP Status: 500 Internal Server Error

```
rpc error: code = FailedPrecondition desc = Can't access cart storage. StackExchange.Redis.RedisConnectionException: It was not possible to connect to the redis server(s). Error connecting right now. To allow this multiplexer to continue retrying until it's able to connect, use abortConnect=false in your connection string or AbortOnConnectFail=false; in your code.
    at StackExchange.Redis.ConnectionMultiplexer.ConnectImplAsync(ConfigurationOptions, TextWriter, Nullable`1) in /_/src/StackExchange.Redis/ConnectionMultiplexer.cs:line 609
    at Microsoft.Extensions.Caching.StackExchangeRedis.RedisCache.ConnectAsync(CancellationToken)
    at Microsoft.Extensions.Caching.StackExchangeRedis.RedisCache.GetAndRefreshAsync(String, Boolean, CancellationToken)
    at Microsoft.Extensions.Caching.StackExchangeRedis.RedisCache.GetAsync(String, CancellationToken)
    at cartservice.cartstore.RedisCartStore.GetCartAsync(String) in /app/cartstore/RedisCartStore.cs:line 90
could not retrieve cart
```

Found it, the service for redis-cart is called redis-cartv -> typo
-> kubectl get svc

```
192:Module_10_chapter-21 astrid$ kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
adservice	ClusterIP	10.128.174.215	<none>	9555/TCP	21m
cartservice	ClusterIP	10.128.35.6	<none>	7070/TCP	21m
checkoutservice	ClusterIP	10.128.168.110	<none>	5050/TCP	21m
currencyservice	ClusterIP	10.128.65.66	<none>	7000/TCP	21m
emailservice	ClusterIP	10.128.211.241	<none>	5000/TCP	21m
frontend	LoadBalancer	10.128.75.79	172.236.5.79	80:31083/TCP	21m
kubernetes	ClusterIP	10.128.0.1	<none>	443/TCP	80m
paymentservice	ClusterIP	10.128.143.156	<none>	50051/TCP	21m
productcatalogservice	ClusterIP	10.128.84.160	<none>	3550/TCP	21m
recommendationservice	ClusterIP	10.128.171.55	<none>	8080/TCP	21m
redis-cartv	ClusterIP	10.128.228.76	<none>	6379/TCP	21m
shippingservice	ClusterIP	10.128.95.8	<none>	50051/TCP	21m

And here it is:

```
Module_10_chapter-21 > charts > redis > templates > service.yaml
```

```
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: {{ .Values.appName }}v
5  spec:
6    type: ClusterIP
```

So I remove the 'v' and try again
-> helmfile sync

Helmfile will:

- └─ detect changes in your templates ✓
- └─ upgrade all releases automatically ✓
- └─ apply the fixed service.yaml ✓


```

NAME: currencyservice
LAST DEPLOYED: Fri May 22 11:17:01 2026
NAMESPACE: default
STATUS: deployed
REVISION: 2
DESCRIPTION: Upgrade complete
TEST SUITE: None

Listing releases matching ^currencyservice$
cartservice    default    2          2026-05-22 11:17:01.216536 +0100 BST    deployed    microservice-0.1.0    1.16.0

Release "paymentservice" has been upgraded. Happy Helming!
NAME: paymentservice
LAST DEPLOYED: Fri May 22 11:17:01 2026
NAMESPACE: default
STATUS: deployed

```

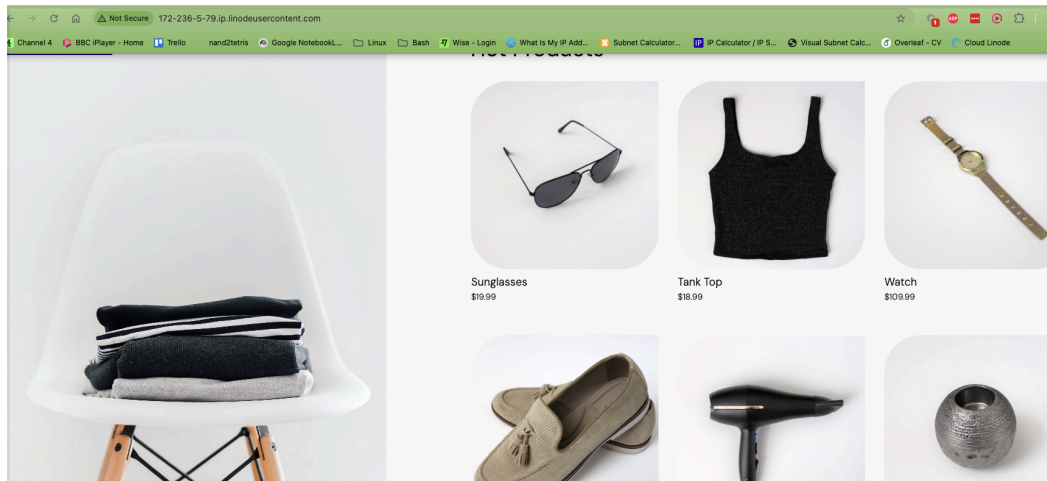
The redisservice has the correct name now so CartService can use it:

```

192:Module_10_chapter-21 astrid$ kubectl get svc
NAME                                TYPE                CLUSTER-IP          EXT
adservice                          ClusterIP            10.128.174.215      <no
cartservice                        ClusterIP            10.128.35.6         <no
checkoutservice                   ClusterIP            10.128.168.110      <no
currencyservice                   ClusterIP            10.128.65.66        <no
emailservice                      ClusterIP            10.128.211.241      <no
frontend                          LoadBalancer        10.128.75.79        172
kubernetes                        ClusterIP            10.128.0.1          <no
paymentservice                   ClusterIP            10.128.143.156      <no
productcatalogservice            ClusterIP            10.128.84.160       <no
recommendationservice            ClusterIP            10.128.171.55       <no
redis-cart                        ClusterIP            10.128.81.104       <no
shippingservice                  ClusterIP            10.128.95.8         <no

```

And it works 🥳



Uninstall releases

-> helmfile destroy

It looks at the current state and starts taking action to get to the desired state:

```
192:Module_10_chapter-21 astrid$ helmfile destroy
Building dependency release=recommendationservice, chart=charts/microservice
Building dependency release=paymentservice, chart=charts/microservice
Building dependency release=shippingservice, chart=charts/microservice
Building dependency release=productcatalogservice, chart=charts/microservice
Building dependency release=emailservice, chart=charts/microservice
Building dependency release=checkoutservice, chart=charts/microservice
Building dependency release=frontend-service, chart=charts/microservice
Building dependency release=currencyservice, chart=charts/microservice
Building dependency release=cartservice, chart=charts/microservice
Building dependency release=ad-service, chart=charts/microservice
Building dependency release=redis-cart, chart=charts/redis
Listing releases matching ^shippingservice$
shippingservice default      2      2026-05-22 11:17:01.301743 +0100 BST    deployed      microservice-0.1.0      1.16

Listing releases matching ^recommendationservice$
recommendationservice default      2      2026-05-22 11:17:01.329592 +0100 BST    deployed      microservice-0.1.0
.16.0
```

```
Deleting emailservice
Deleting currencyservice
Deleting shippingservice
Deleting checkoutservice
Deleting adservice
Deleting cartservice
Deleting redis
Deleting recommendationservice
Deleting paymentservice
Deleting productcatalogservice
Deleting frontendservice
release "recommendationservice" uninstalled

release "paymentservice" uninstalled

release "shippingservice" uninstalled

release "emailservice" uninstalled

release "checkoutservice" uninstalled
```

DELETED RELEASES:

NAME	NAMESPACE	DURATION
recommendationservice		1s
paymentservice		1s
shippingservice		1s
emailservice		1s
checkoutservice		1s
frontendservice		1s
adservice		2s
productcatalogservice		2s
cartservice		2s
redis-cart		2s
currencyservice		2s

Now we check for releases:

-> helm list

```
192:Module_10_chapter-21 astrid$ helm list
```

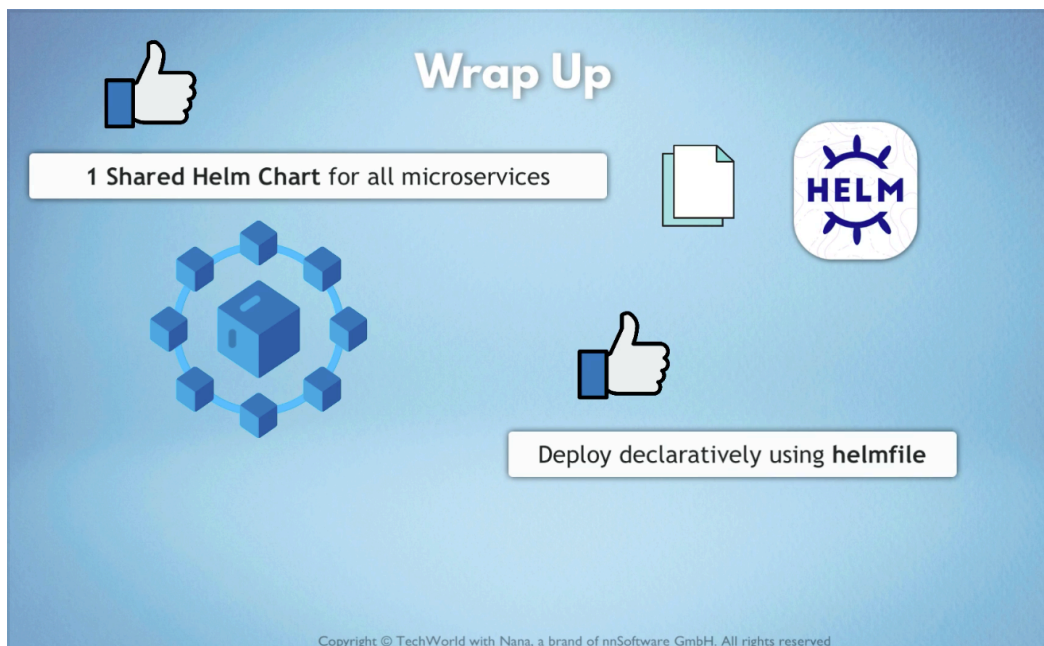
NAME	NAMESPACE	REVISION	UPDATED	STATUS	CHART	APP VERSION
------	-----------	----------	---------	--------	-------	-------------

And for pods:

-> kubectl get pod

```
192:Module_10_chapter-21 astrid$ kubectl get pod
```

No resources found in default namespace.

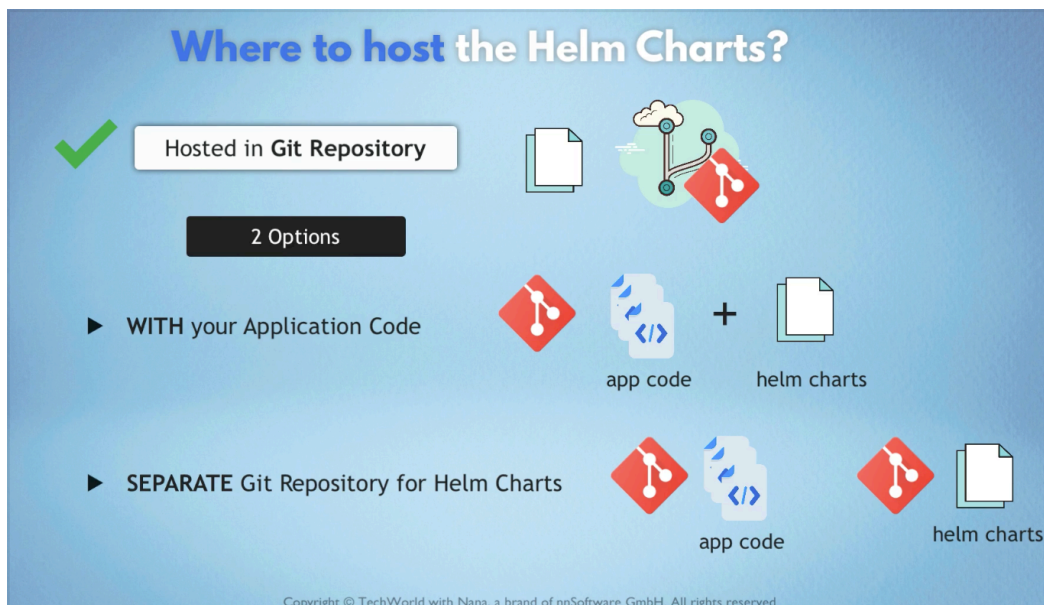


Where to host the helm charts?

-> git repository

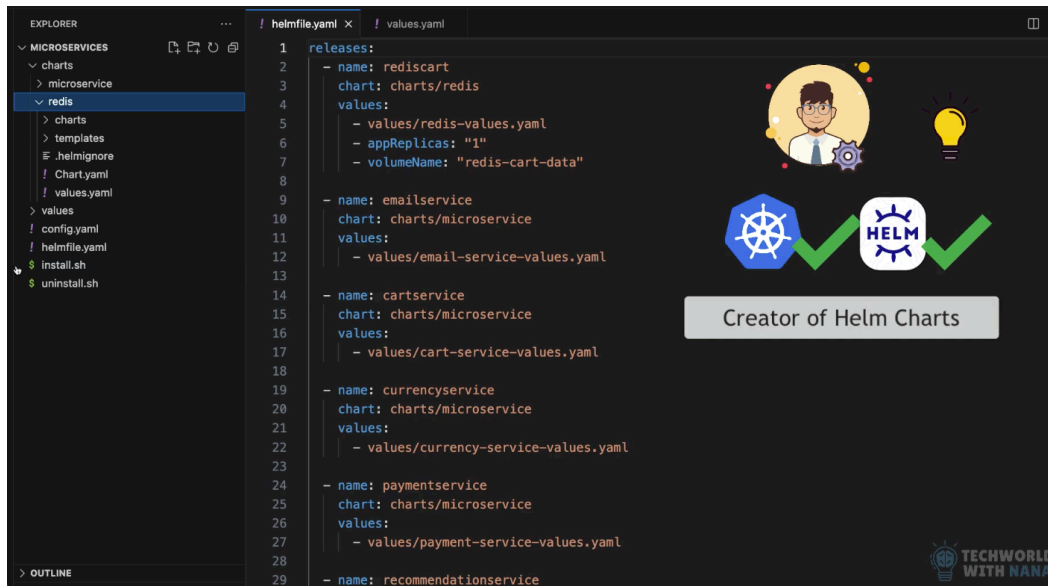
Two options:

1. Helm charts are with our app code
2. Preferred way: separate git repo only for helm charts

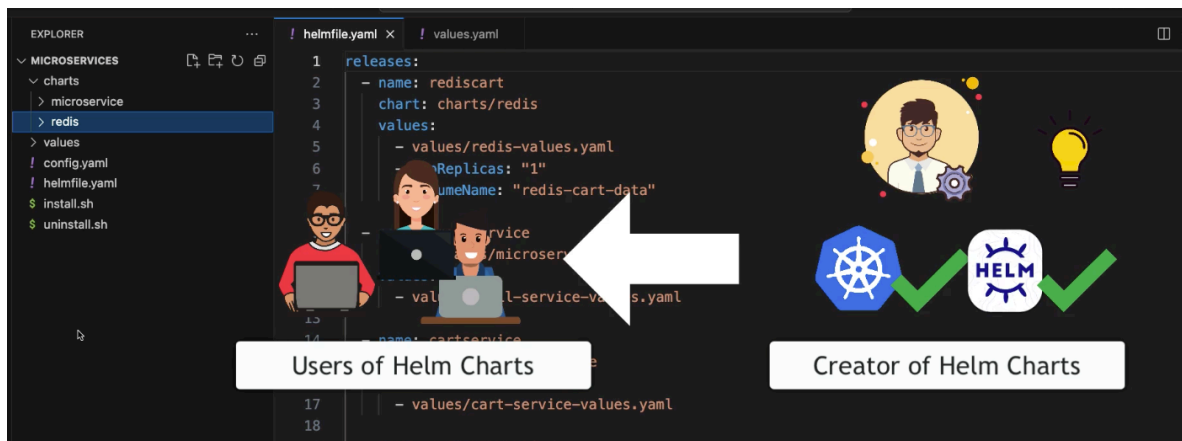


How does it fit into the DevOps workflow (CI/CD pipeline)?

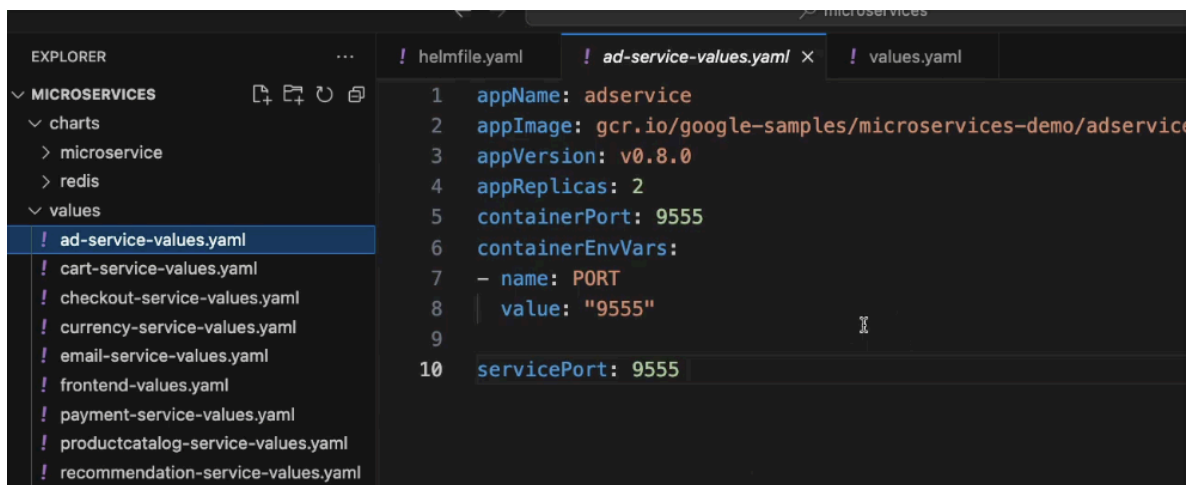
The DevOps Engineer creates the helm charts:



And the application developers will be the users of the helm charts



And the users can make changes to the values:



And the users then can use the charts to deploy their MS application into the cluster usually by integrating them into the CI/CD pipeline.



Facilitators for Developer & Operations Team

- ▶ Creating configurations and programs for these teams

devOps facilities -> users can optimise their own workflows

In CI/CD pipeline we execute the helmfile from the build automation tool